

Prakshit Mathur

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SUMMARY

I am a passionate ML enthusiast and 3rd-year CS undergraduate (CGPA 8.55/10, BIT Mesra) who genuinely loves exploring how machines learn and make decisions. Driven by curiosity, I built end-to-end AI/ML projects across NLP, LLMs, RAG, and generative deep learning. Oracle Data Science certified and I am eager to bring hands-on energy to a team where I can learn fast, contribute faster, and grow.

PROJECTS

- RAG-Based Intelligent Document Assistant** Python · LLMs · Embeddings · FAISS · FastAPI [Link](#) 2026
- Independently architected a complete RAG pipeline — document ingestion → chunk embeddings → FAISS vector-store → context-gated LLM response.
 - Implemented **cosine similarity thresholding** on retrieved chunks to filter low-confidence context before LLM inference, reducing off-topic responses.
- AI Research Paper Analyzer** Python · RAG · LLM · FastAPI [GitHub](#) Ongoing
- Designed an autonomous paper discovery pipeline that scrapes ArXiv, Semantic Scholar, and Papers With Code every 6 hours using APScheduler and REST APIs, auto-ranking top 10 papers by relevance score
 - Built a semantic search interface supporting natural language Q&A grounded in the paper corpus, with BibTeX export and side-by-side paper comparison
- Text-to-Music Generation using Transformers** Python · PyTorch · TransformerArchitecture [Link](#) 2025
- Developed an AI-powered Text-to-Music generation system using deep learning and generative AI to convert text prompts into music through automated audio synthesis.
- SMS Spam Detection — Production NLP Service** Python · TF-IDF · MultinomialNaive Bayes · FastAPI [Link](#) 2024
- Built an end-to-end NLP pipeline — preprocessing, TF-IDF vectorization, Naive Bayes training — achieving **92% accuracy** on class-imbalanced data; packaged into a reproducible FastAPI inference service with confidence scoring.

EXPERIENCE

Machine Learning Trainee

2025

- Built a cancer-prediction classification pipeline achieving **88% CV score** — executed the full ML lifecycle: data cleaning, EDA, feature engineering, hyperparameter tuning, and evaluation reporting.

Python & ML Intern — YBI Foundation

2024

- Applied regression and classification algorithms on structured datasets; delivered documented model comparison reports covering accuracy, precision, recall, and F1 score.

TECHNICAL SKILLS

Languages: Python, C++

AI / ML: NLP, retrieval-augmented generation (RAG), transformer-based models, classical ML

Libraries: PyTorch, Scikit-learn, Pandas, NumPy, FastAPI, OpenCV, Matplotlib, Github, Streamlit

Concepts: Vector Stores, Embeddings, Semantic Search, OOP, DSA, Data Preprocessing

EDUCATION

B.Tech in Computer Science & Engineering — Birla Institute of Technology, Mesra | 2023–2027 | CGPA: **8.55 / 10**

ACHIEVEMENTS & CERTIFICATIONS

- Oracle Data Science Professional Certified** — validated end-to-end ML lifecycle competency including model building, evaluation, and deployment. [Proof](#)
- 150+ LeetCode problems solved** — demonstrates strong algorithmic problem-solving under timed competitive conditions.
- 5-Star HackerRank rating in C++** — in the top percentile for language proficiency, active rated competitor on CodeChef and Codeforces.